

Reactive oxygen species assessment

Material:

1. DCF-DA stock solution (final concentration 10 mM)
2. DMSO
3. TRIS-HCl buffer solution
4. 96 well plate

Procedures:

1. Immediately prior to use, prepare a fresh stock solution of DCF-DA in sterile dimethylsulfoxide (DMSO) at a final concentration of 10mM.
2. Collect the spinal cord fragment.
3. Cool immediately the fragment to -20 °C in TRIS-HCl buffer solution (10 mM, pH 7.4).
4. After homogenization, centrifuge the homogenate (10,000 rpm, 4 °C, 10 min).
5. Add a 20-μL aliquot of this supernatant to 80 μL of DCF-DA (final concentration 125 μM).
6. Incubate triplicate plates in the dark at 37 °C for 1 h.
7. Assess ROS by immediately analyzing your cells by measure the fluorescence (488/525 nm) in a Victor X4 apparatus (Perkin-Elmer).
8. Normalize the results by protein content.